

Precipitation

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- 1 Guest lecture next class on Snow/Snowmelt
- 2 Problem set #2 due tonight

Learning Objectives

- 1 Requirements for precipitation to occur.
- 2 Uplift mechanisms
- 3 Measuring rain (and associated challenges)
- 4 Areal estimation techniques
- 5 Rainfall statistics

Areal Estimation

Hydrologists are often interested in precipitation over a *region* (rather than discrete points), for water budget calculations and hydrologic response analysis.

These can largely be categorized as:

- 1 Weighted averages: estimating a single P for a region.
- 2 Surface-fitting methods: estimating a continuous field of precipitation.

Areal Estimation

The general form for the estimation of precipitation ($\hat{P}$) over an area is:

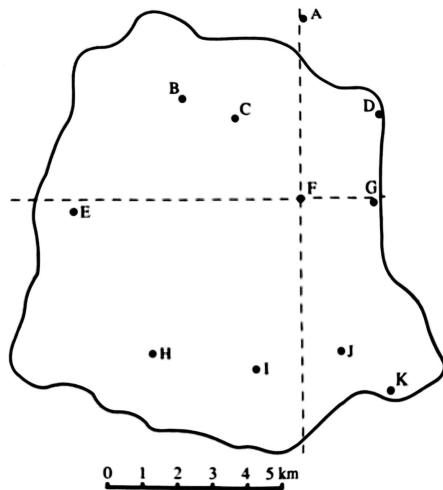
$$\hat{P} = \frac{1}{A} \cdot \int \int p(x, y) \cdot dx \cdot dy$$

where P is the average precipitation over a region, A is the area of the region, x, y are the coordinates of point measurements, and $p(x, y)$ is the precipitation at those points.

Weighted Averages

The simplest is an **arithmetic average**, which assumes equal weight for all gages equal to $\frac{1}{G}$:

$$\hat{P} = \frac{1}{G} \sum_{g=1}^G p_g$$

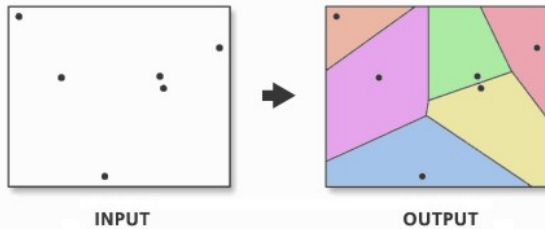


Weighted Averages

Slightly more sophisticated is an area-based weighting approach called **Thiessen Polygons**:

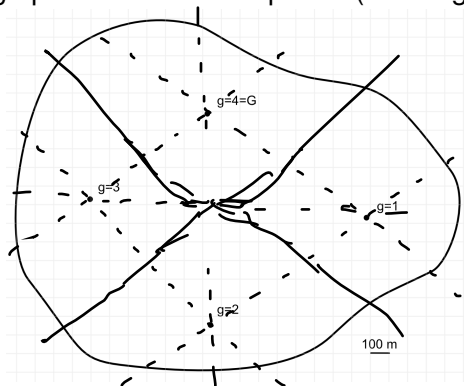
$$\hat{P} = \frac{1}{A} \sum_{g=1}^G a_g \cdot p_g$$

where A is total area of the region and a_g is the area of the polygon.



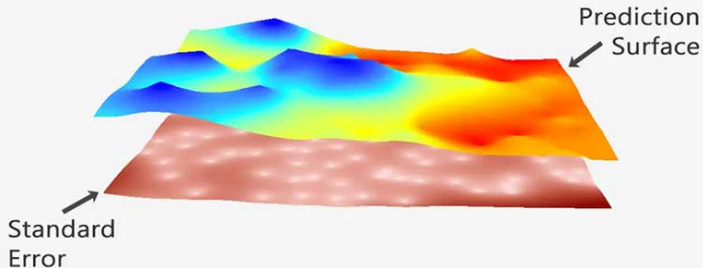
Thiessen Polygons demo

- 1 Draw straight line connecting adjacent gages, forming a network of triangles.
- 2 Perpendicular bisectors are drawn on each line until they connect, forming polygons around the gages.
- 3 The area of these polygons can be estimated by overlaying the network and region on a grid and counting up the number of squares (or using GIS).



Surface-Fitting Methods

Measured values are used to identify a surface that represents precipitation at all data points in a region of interest. With **smoothing**, the surface doesn't have to exactly honor the point data, where **interpolation** honors the point data.



Inverse-Distance Weighting

$$D = \sum_{g=1}^G d_g^{-b}$$

$$\hat{p}_0 = \frac{1}{D} \cdot \sum_{g=1}^G d_g^{-b} p_g$$

where d is distance from gage g to the unmeasured point and b is a power coefficient

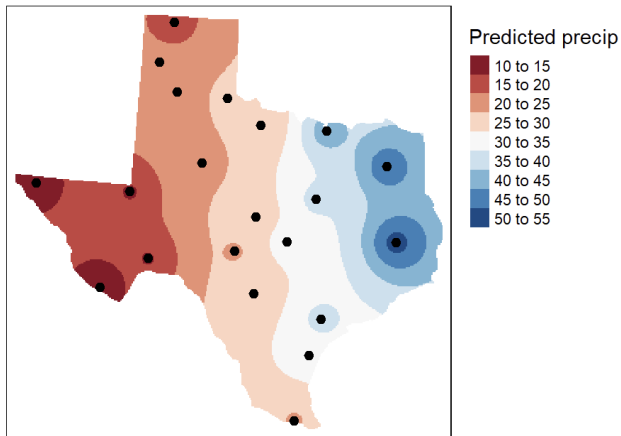
that expresses how strong we want to weight nearby points (1 being the least and 2 being a common value). This equation is "swept" over a grid of unmeasured precipitation points, a trivial operation using GIS software/packages.

A Really Nice Tutorial on Spatial Interpolation

This tutorial walks through Thiessen polygon, IDW, and Kriging (with de-trending) to interpolate precipitation in TX

<https://mgimond.github.io/Spatial/spatial-interpolation.html>

IDW interpolation



Limitations:

- Bullseye effect for outliers
- Biased towards clustered data
- only consider distance

Hypsometric Method

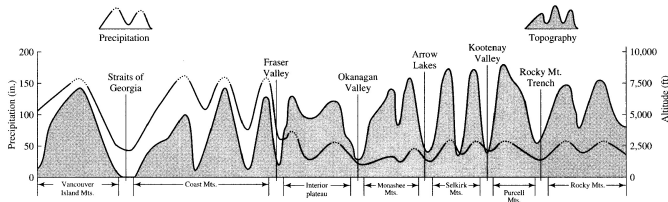


Figure 4.14 Relation between mean annual precipitation and topography across southern British Columbia, Canada [adapted from Bruce and Clark (1966)].

We saw in the last lecture that precipitation varies with elevation and that relationship can be empirically determined to form a **lapse rate**.

This lapse is the basis of the hypsometric method. A relationship is determined between precip (p) and elevation (z), often using a linear regression:

$$p(z) = a + b \cdot z$$

PRISM

Parameter-elevation Regressions on Independent Slopes Model (PRISM) is an example of an algorithmic hypsometric method—it primarily relies on the regression with elevation, but also takes into account aspect, coastal proximity, and topographic index.

Statistics incoming!



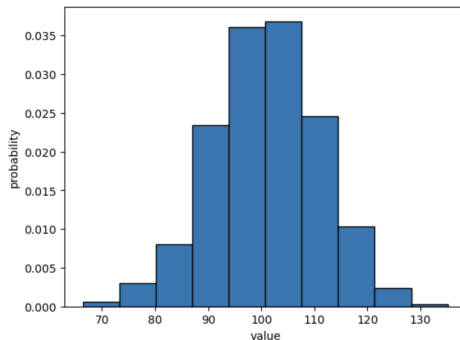
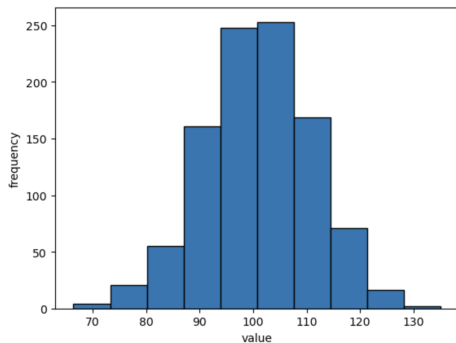
Frequentist Inference

The **probability** of an event is equivalent to the **frequency** of an event. Underpins much of the statistical analysis in hydrology including hypothesis testing and confidence intervals.

Frequency is determined by repeated tests of a controlled experiment, effectively sampling the underlying **population**.

The statistical properties of the population, mean (μ) and variance (σ^2) are assumed to be fixed, unknown quantities. This is referred to as **stationarity**.

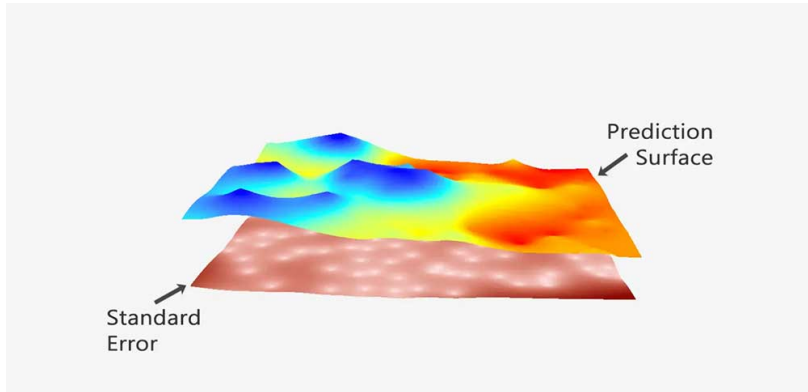
Frequentist Inference



1000 draws from a normal distribution with mean (μ) = 100 and standard deviation (σ) = 10.

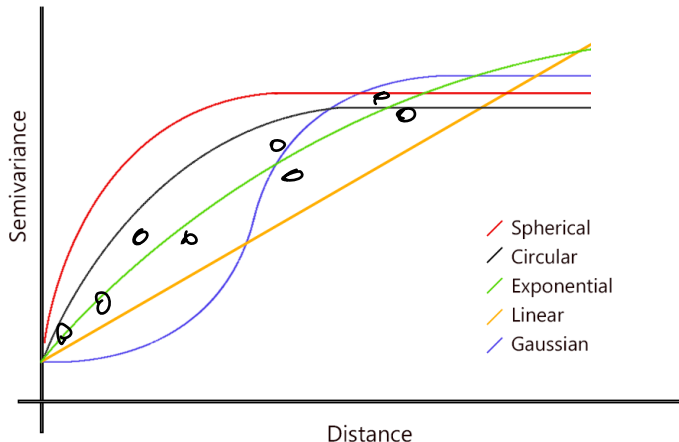
Kriging Interpolation

Kriging is a statistical method which uses spatial auto-correlation to calculate the weights of point data to interpolate values for the region of interest. Advantages over other interpolation methods are that it is not biased by clustered data and provides an estimate of interpolation uncertainty for the entire field.



Spatial Auto-Correlation

The interpolation weights used in kriging rely on a characterization of the spatial auto-correlation of the variable (in this case, precipitation), called a **variogram**.



Kriging Assumptions

Ordinary kriging (OK) assumes isotropy, or uniformity in all directions, and stationarity. If either are not satisfied, options are:

- Remove the trend, krig the residuals, and then add trend back in
- Use a kriging method that allows for a non-stationary trend.
Universal Kriging relaxes the stationarity assumption, allowing the mean to change but variance to be constant in space. This is "weak stationarity."
- Fit multiple "directional" variograms or perform "block kriging" (krige smaller scales that are isotropic to account for anisotropy).

Kriging Approach

- 1 Fit variogram model to data.
- 2 Calculate kriging factors (interpolation weights).
- 3 Interpolate values to the continuous field based on the kriging factors.

<https://mgimond.github.io/Spatial/spatial-interpolation.html>

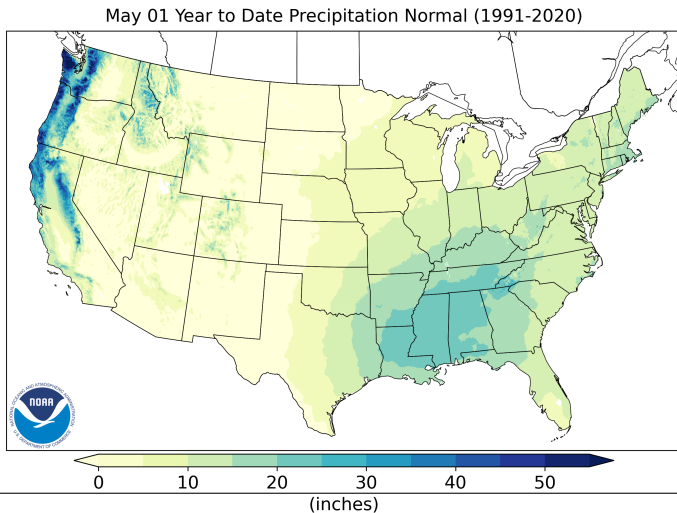
Precipitation Climatology

Long-term average precipitation rates are fundamental determinants of regional hydrology and vegetation, whereas variability and extreme values have important consequences for human livelihoods and infrastructure.

- 1 Long-term average (30-year normals)
- 2 Probable maximum precipitation
- 3 Depth-duration frequency analysis

Long-term Average

National Centers for Environmental Information generate 30-year climate normals every ten years.



Probable Maximum Precipitation (PMP)

PMP is the theoretical maximum depth of precipitation for a given storm over a given duration in a given area at a given time of year.

Engineers use PMP to estimate the **probable maximum flood** (PMF) from which to design large, critical facilities like dams, spillways, or nuclear power plants to address high hazard risks.



PMP Approach

The Hershfield method is a statistical method to estimate PMP provided there is a sufficiently long time series

$$X_{PMP} = \mu_N + K \cdot \sigma_N$$

where X_{PMP} is the probable maximum precipitation, μ_N is the mean of the annual maximum precipitation time series for a given storm duration, σ_N is the standard deviation of that time series, and K is frequency factor determined by:

$$K = \frac{X_M - \mu_{N-1}}{\sigma_{N-1}}$$

where X_M is the maximum value in the series of at the location, and μ_{N-1} and σ_{N-1} are the mean and standard deviation of the time series, respectively for (N-1) years after removing the year with the maximum value.

PMP Mapper

NM-CO PMP mapper

Frequency Analysis

Most applications don't require PMP-level designs. **Frequency analysis** is more appropriate in less extreme circumstances.

The underlying assumption for frequency analysis is that there is a sufficiently long record that captures the “full” distribution and that the moments of that distribution do not change over time (i.e., the stationarity assumption).

Two approaches exist for rainfall:

- 1 depth-duration frequency (DDF) analysis
- 2 intensity-duration frequency (IDF) analysis (depth divided by time = intensity)

Depth-duration Frequency Analysis

Goal: estimate probability that a given rainfall amount (for a specified duration) will be exceeded in a given year.

Graphical Depth-duration Frequency Analysis

- 1 Collect (or download) data and calculate annual maximum amounts for specific durations (1-hr, 2-hr, 24-hr, etc.)
- 2 Sort annual values from largest to smallest for given duration
- 3 Calculate exceedance probability using Weibull plotting position

$$P = \frac{m}{N + 1}$$

where P is annual exceedance probability, m is the rank from 1 (largest depth) to N (number of years)

Return period (or interval) in years is the inverse of P

$$R = \frac{1}{P}$$

- 4 Plot log depth by log probability and "fit" (by hand) line through points.

Limitations of Frequency Analysis

- Weibull distribution not great for extremes (others are better)
- Sensitivity to sample size (true of all frequency approaches)
- Stationarity assumption violated by climate change

To this end, NOAA Atlas 14 (precipitation frequency over CONUS) is currently being updated (NOAA Atlas 15) to account for climate change.